StuTech Science Notes

Basic Physics Formula Sheet

A quick reference for introductory science

Learning goals

- Recognize common introductory physics formulas.
- Match physical quantities with SI units.
- Practice substituting values into equations.

1. Motion

Quantity	Formula	Meaning
Speed	speed = distance ÷ time	How much distance is covered per unit time.
Average velocity	displacement ÷ time	Rate of change of position in a direction.
Acceleration	a = (v – u) ÷ t	Change in velocity per unit time.

2. Forces and energy

Quantity	Formula	Unit
Force	F = m × a	newton (N)
Weight	W = m × g	newton (N)
Work	work = force × distance (in direction of force)	joule (J)
Power	P = energy transferred ÷ time	watt (W)
Kinetic energy	KE = ½mv²	joule (J)
Gravitational potential energy near Earth	GPE = mgh	joule (J)

3. Density, pressure and electricity

Quantity	Formula	Unit
Density	ρ = mass ÷ volume	kg/m³
Pressure	p = force ÷ area	pascal (Pa)
Electric charge	Q = I × t	coulomb (C)
Voltage	V = energy transferred ÷ charge	volt (V)
Ohm’s law	V = I × R	volt (V)

4. Waves

Wave speed = frequency × wavelength ($v = f\lambda$). Frequency is measured in hertz (Hz), wavelength in metres (m), and wave speed in metres per second (m/s).

5. Important reminders

• Convert values to consistent units before calculating.

• Write the formula, substitute values, then calculate.

• Include units in the final answer.

• Check whether the question asks for a scalar quantity (such as speed) or a vector (such as velocity).

• These equations are a revision aid; your course may use additional conditions or notation.

6. Practice problems

1. A cyclist travels 120 m in 20 s. Find the average speed.

2. A 3 kg object accelerates at 2 m/s². Find the net force.

3. A device transfers 600 J of energy in 30 s. Find its average power.

4. A wave has frequency 5 Hz and wavelength 2 m. Find its speed.

Study tip: Read each section, write a short summary in your own words, then answer the review questions without looking at the notes.

Educational resource prepared for the StuTech demo library. Check your course syllabus and teacher guidance for examinable detail.